

Notes: Wu & Zhang (RFS, 2025)
Common Pricing of Decentralized Risk: A Linear Option Pricing Model

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1 Big picture

- **Question.** Can option prices be unified without specifying a full bottom-up risk-neutral dynamics for the underlying asset?
- **Traditional benchmark.** Black–Scholes–Merton-style models start from an underlying price process and value all options from that common dynamics.
- **Problem with bottom-up models.** A single dynamics specification often struggles to match the full option surface and can become disconnected from the realized risk behavior used in risk management.
- **Paper’s approach.** The authors propose a top-down linear option pricing model. Each option is decomposed into BMS risk sensitivities, and pricing is centralized by imposing a common market price per unit of each decentralized risk estimate.
- **Key distinction.** The model does *not* assume common risk dynamics across options. It allows each contract to have its own historical risk estimates, then requires the market price of each risk source to be common across contracts.
- **Risk sources.** The four main risk dimensions are vega, gamma, volga, and vanna. These correspond to implied-volatility level/term structure, underlying return variance, implied-volatility variance/smile curvature, and return–volatility covariance/skew.
- **Empirical setting.** The application uses OTC currency options on USD/JPY and USD/GBP from 1996 to 2022, with 25 implied-volatility quotes for each currency on a grid of five maturities and five put-delta strikes.
- **Main pricing result.** The model explains option time-decay cross sections extremely well. Median cross-sectional pricing-regression R^2 values are about 99.9% for both currencies.
- **Main error result.** The average implied-volatility pricing errors are about 22 basis points for pound options and 27 basis points for yen options, less than half those of common bottom-up benchmarks discussed in the paper.
- **Return-factor result.** The same risk-adjusted exposures also explain realized delta-hedged option returns. The model-implied option return factor structure has average R^2 around 88–89%.
- **Predictability result.** The estimated market price of risk for a risk dimension predicts future excess returns of the corresponding risk-targeting portfolio, with slope estimates close to one.
- **Investment result.** Timing risk-targeting portfolios with the model-implied market price of risk improves information ratios relative to fixed notional short-option strategies.
- **Economic takeaway.** The paper links option pricing, risk management, and option return attribution under a single linear framework. Fair pricing comes from common pricing of decentralized risk, not from a fully specified underlying dynamics.

2 Data and measurement

2.1 Option sample

- The empirical analysis uses OTC currency options on the dollar price of Japanese yen and British pound.
- The sample runs from 1996 to 2022.
- For each currency and date, the option surface consists of 25 implied-volatility quotes.
- The grid has five maturities: 1, 3, 6, 9, and 12 months.
- The grid has five absolute put deltas: 10, 25, 50, 75, and 90.
- The authors estimate the pricing model every Wednesday from January 29, 1997 to October 12, 2022, after using the first year of data to form historical moment estimators.

Interpretation: The data are ideal for the model because the option surface is standardized, repeated over a long time series, and has enough cross-sectional contracts to identify multiple risk-price coefficients at each date.

2.2 Why BMS implied volatility is used

- The paper uses the BMS formula as a *representation* of observed option values, not as a belief that the true underlying dynamics are geometric Brownian motion.
- Every option price is converted into an implied volatility $I_{t,i}$.
- The BMS value function $B^i(t, S_{t,i}, I_{t,i})$ summarizes the option value using observable time, forward price, and implied volatility.
- BMS sensitivities provide the common language for risk decomposition: theta, delta, vega, gamma, volga, and vanna.

Interpretation: The BMS formula is a measurement device. It transforms heterogeneous option prices into comparable risk exposures.

2.3 Historical moment estimators

For each option contract i , the paper estimates:

- historical forward-return variance $\hat{v}_{t,i}$;
- historical proportional implied-volatility variance $\hat{\omega}_{t,i}$;
- an average return-implied-volatility correlation estimate $\hat{\rho}_t$ for vanna-risk null values.

Interpretation: The key empirical innovation is that these moment estimates are decentralized. Different options can have different risk magnitudes, but each risk source is priced by a common proportionality coefficient.

2.4 Delta-hedged excess returns

The option-return tests focus on weekly delta-hedged excess returns from writing options. Delta hedging removes first-order exposure to the underlying forward price, leaving the return to be explained by vega, gamma, volga, vanna, and pricing-error reversal.

Interpretation: The same exposures used to price time decay are then used to explain and predict realized option returns. This is what integrates pricing with risk management.

3 Models and empirical specifications

3.1 Top-down P&L attribution

For option i , write its BMS value representation as

$$B^i(t, S_{t,i}, I_{t,i}),$$

where $S_{t,i}$ is the underlying forward price and $I_{t,i}$ is the option's BMS implied volatility. A second-order Taylor expansion gives

$$dB^i = B_t^i dt + B_S^i dS_{t,i} + B_I^i dI_{t,i} \quad (1)$$

$$+ \frac{1}{2} B_{SS}^i (dS_{t,i})^2 + \frac{1}{2} B_{II}^i (dI_{t,i})^2 + B_{IS}^i dS_{t,i} dI_{t,i} + \dots \quad (2)$$

Interpretation: The option's P&L is decomposed into time decay plus exposures to price and implied-volatility movements. The second-order terms are crucial because delta-hedged option returns are dominated by curvature exposures.

3.2 Statistical expected excess-return attribution

Under the statistical measure P , the proportional moments are

$$\begin{aligned} \phi_{t,i}^P &= \frac{1}{dt} E_t^P \left[\frac{dS_{t,i}}{S_{t,i}} \right], \quad \mu_{t,i}^P = \frac{1}{dt} E_t^P \left[\frac{dI_{t,i}}{I_{t,i}} \right], \\ v_{t,i}^P &= \frac{1}{dt} E_t^P \left[\left(\frac{dS_{t,i}}{S_{t,i}} \right)^2 \right], \quad \omega_{t,i}^P = \frac{1}{dt} E_t^P \left[\left(\frac{dI_{t,i}}{I_{t,i}} \right)^2 \right], \\ \gamma_{t,i}^P &= \frac{1}{dt} E_t^P \left[\frac{dS_{t,i}}{S_{t,i}} \frac{dI_{t,i}}{I_{t,i}} \right]. \end{aligned}$$

After delta hedging, the risk contributions are

$$\text{PRC}_{t,i} = \left(B_I^i I_{t,i} \mu_{t,i}^P, \frac{1}{2} B_{SS}^i S_{t,i}^2 v_{t,i}^P, \frac{1}{2} B_{II}^i I_{t,i}^2 \omega_{t,i}^P, B_{IS}^i I_{t,i} S_{t,i} \gamma_{t,i}^P \right).$$

Interpretation: These terms are the break-even contributions from vega, gamma, volga, and vanna risk. If a delta-hedged option has zero expected excess return, these contributions exactly offset time decay.

3.3 Decentralized pricing relation

Under the risk-neutral measure Q , no dynamic arbitrage requires zero expected excess returns for the option and the underlying. The single-contract pricing relation is

$$-B_t^i = B_I^i I_{t,i} \mu_{t,i} + \frac{1}{2} B_{SS}^i S_{t,i}^2 v_{t,i} + \frac{1}{2} B_{II}^i I_{t,i}^2 \omega_{t,i} + B_{IS}^i I_{t,i} S_{t,i} \gamma_{t,i}. \quad (3)$$

Interpretation: This is decentralized because it prices one option using that option's own exposures and risk-neutral moments. By itself, it is too flexible to identify because each contract could have its own moments.

3.4 Common pricing of decentralized risk

The paper imposes common market pricing per unit of estimated risk:

$$\begin{aligned}\mu_{t,i} &= \beta_{V,t}\hat{\omega}_{t,i}, & v_{t,i} &= \beta_{G,t}\hat{v}_{t,i}, & \omega_{t,i} &= \beta_{O,t}\hat{\omega}_{t,i}, \\ \gamma_{t,i} &= \beta_{A,t}\sqrt{\hat{v}_{t,i}\hat{\omega}_{t,i}}.\end{aligned}$$

Plugging these into the decentralized pricing relation gives the linear option pricing model:

$$-B_t^i = \beta_{V,t}\hat{\omega}_{t,i}B_{II}^iI_{t,i} + \frac{1}{2}\beta_{G,t}\hat{v}_{t,i}B_{SS}^iS_{t,i}^2 \quad (4)$$

$$+ \frac{1}{2}\beta_{O,t}\hat{\omega}_{t,i}B_{II}^iI_{t,i}^2 + \beta_{A,t}\sqrt{\hat{v}_{t,i}\hat{\omega}_{t,i}}B_{IS}^iS_{t,i}I_{t,i}. \quad (5)$$

Interpretation: Risk magnitudes are contract-specific, but the price per unit of each risk dimension is common across contracts at a given date. This is the paper’s central identifying assumption.

3.5 Null values and market prices of risk

Under zero risk premium, zero implied-volatility trend predictability, and no historical forecast bias, the null coefficient vector is

$$\beta_t^0 = (0, 1, 1, \hat{\rho}_t)^\top.$$

The market price of risk estimate is defined as

$$\eta_t = \beta_t - \beta_t^0 = (\beta_{V,t}, \beta_{G,t} - 1, \beta_{O,t} - 1, \beta_{A,t} - \hat{\rho}_t)^\top.$$

Interpretation: The phrase “market price of risk” is used broadly. Deviations from the null can reflect risk premia, mispricing, or differences between market expectations and the simple historical moment forecasts.

3.6 Cross-sectional regression identification

At each date t , define

$$y_{t,i} = -B_t^i.$$

The cross-sectional pricing regression is

$$\mathbf{y}_t = X_t \mathbf{b}_t + \mathbf{e}_t, \quad (6)$$

where the i th row of X_t is

$$X_{t,i} = \left(B_{II}^i I_{t,i} \hat{\omega}_{t,i}, \frac{1}{2} B_{SS}^i S_{t,i}^2 \hat{v}_{t,i}, \frac{1}{2} B_{II}^i I_{t,i}^2 \hat{\omega}_{t,i}, B_{IS}^i S_{t,i} I_{t,i} \sqrt{\hat{v}_{t,i} \hat{\omega}_{t,i}} \right).$$

The slope vector is

$$\mathbf{b}_t = (\beta_{V,t}, \beta_{G,t}, \beta_{O,t}, \beta_{A,t})^\top.$$

Interpretation: The regression slopes are common risk-pricing coefficients. The residuals are idiosyncratic deviations from common pricing and can be interpreted as statistical-arbitrage pricing errors.

3.7 Risk-targeting option portfolios

The model also implies risk-targeting portfolios. Let $\mathbf{1}_k$ select risk dimension k . The weight vector for the k th risk-targeting portfolio is

$$\mathbf{w}_{t,k} = X_t (X_t^\top X_t)^{-1} \mathbf{1}_k.$$

The corresponding slope coefficient is

$$b_{t,k} = \mathbf{w}_{t,k}^\top \mathbf{y}_t.$$

Interpretation: These portfolios are long and short options so that they target one unit of a single risk-adjusted exposure while being neutral to other risk dimensions.

3.8 Option return factor model

The return factor model uses the same risk-adjusted exposures:

$$\mathbf{r}_{t+1} = X_t \mathbf{f}_{t+1} + \mathbf{e}_t \phi_{t+1} + \boldsymbol{\varepsilon}_{t+1}, \quad (7)$$

where $\mathbf{r}_{t+1}$ is the vector of weekly delta-hedged excess returns from writing options.

Interpretation: The slope coefficients $\mathbf{f}_{t+1}$ are realized excess returns on risk-targeting factor portfolios. The coefficient ϕ_{t+1} captures return from pricing-error reversal.

3.9 Forecasting risk-targeting portfolio returns

The forecasting test asks whether the estimated market price of risk predicts the future return on the matching risk-targeting portfolio:

$$f_{t+1,k} = a_k + b_k \eta_{t,k} + u_{t+1,k}.$$

Interpretation: If the market-price-of-risk estimate is a meaningful ex ante risk premium signal, then a_k should be near zero and b_k should be near one.

4 Identification and empirical design

4.1 What identifies the pricing model

- At each date, there are 25 option contracts per currency.
- The model has four common slope coefficients: vega, gamma, volga, and vanna risk prices.
- BMS sensitivities and historical moment estimators create cross-sectional variation in the risk-adjusted exposure matrix X_t .
- A weekly cross-sectional regression identifies the common risk-pricing coefficients.

Interpretation: Identification comes from requiring all 25 option contracts to price the same risk source consistently in proportion to their contract-specific estimated risk magnitudes.

4.2 Why this differs from bottom-up identification

- Bottom-up models impose a risk-neutral process for the underlying price and perhaps volatility.
- The proposed model does not specify the underlying process.
- It uses historical risk estimates to recognize that different contracts have different risk behavior.
- It imposes common pricing, not common dynamics.

Interpretation: The paper moves the commonality assumption from risk dynamics to risk prices. This avoids forcing all contracts to share the same implied-volatility dynamics.

4.3 What the pricing residuals mean

- The residual $e_{t,i}$ measures contract-specific deviation from common pricing.
- A positive residual means the option's time decay is high relative to what common risk pricing predicts.
- In principle, long cheap contracts and short expensive contracts can form statistical-arbitrage portfolios.
- Residuals are not strict arbitrage under a structural model because the paper does not impose full underlying dynamics.

Interpretation: The residual is a pricing-error measure relative to the paper's common-pricing benchmark, not an arbitrage certificate in a fully specified model.

4.4 Strengths and limitations

- **Strength:** The model is simple and high dimensional enough to fit the option surface well.
- **Strength:** The same exposures explain realized option returns, making pricing and risk management consistent.
- **Strength:** Estimated market prices of risk predict matching risk-targeting portfolio returns.
- **Limitation:** The model does not provide a structural theory for the sign or size of each risk price.
- **Limitation:** Historical moment estimators may be biased forecasts of market expectations.
- **Limitation:** The model is tested on standardized OTC currency option surfaces; applications to less liquid or less standardized option markets may need additional forecasting inputs.

Interpretation: The model is best understood as a top-down empirical asset-pricing framework for options, not as a replacement for all structural derivative-pricing models.

5 Tables and figures

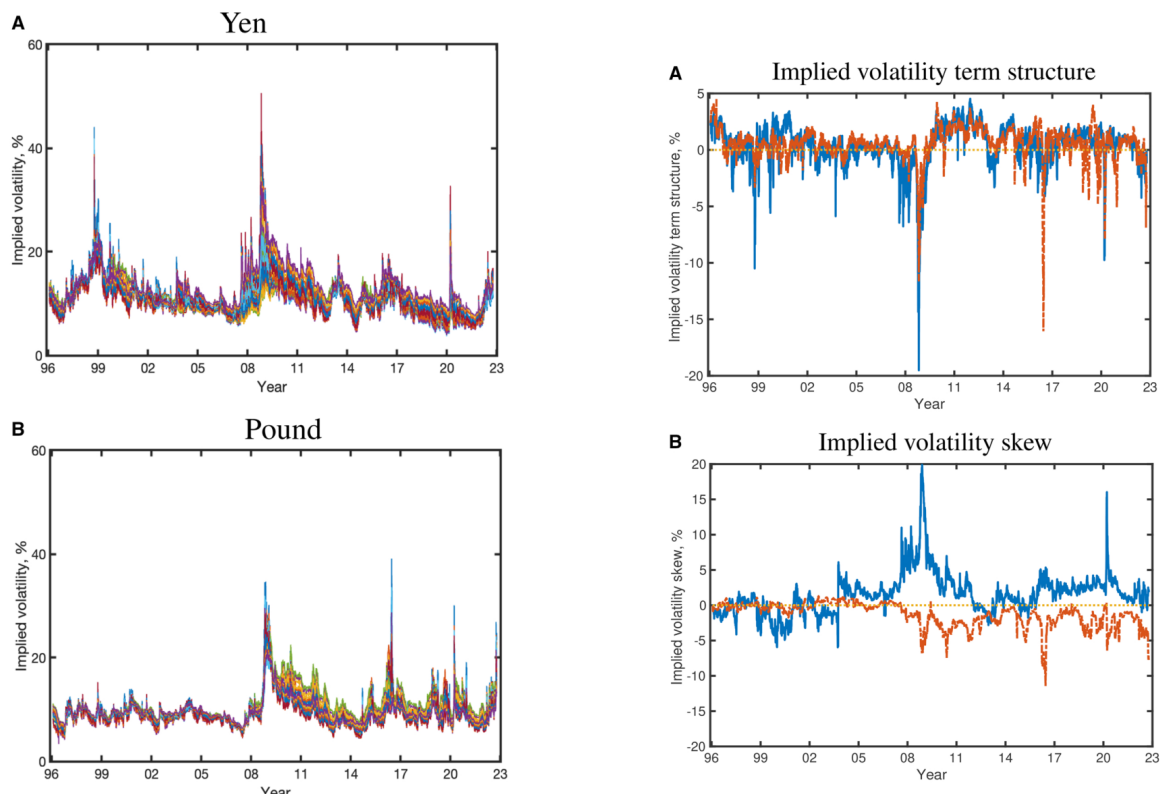
This section follows the same *Read* plus *Economic message* format. Adjacent figures and tables are grouped to save space and speed up comparison. All visuals are directly cropped from the uploaded paper.

5.1 Figures 1–2: implied-volatility level, term structure, and skew

Read:

- Figure 1 plots the full option-surface implied-volatility quotes for yen and pound options.
- Both currencies show large volatility spikes around crisis periods, especially the 2008 financial crisis and later stress episodes.
- Figure 2 decomposes the surface into term-structure and skew measures.
- The term-structure measure often turns negative during crises because short-maturity implied volatility spikes more than long-maturity implied volatility.
- Yen and pound skews diverge substantially in the second half of the sample.

Economic message: Option contracts at different maturities and deltas clearly have different risk behavior. This motivates decentralizing risk estimates rather than imposing common volatility dynamics.



(a) Figure 1: implied-volatility quotes

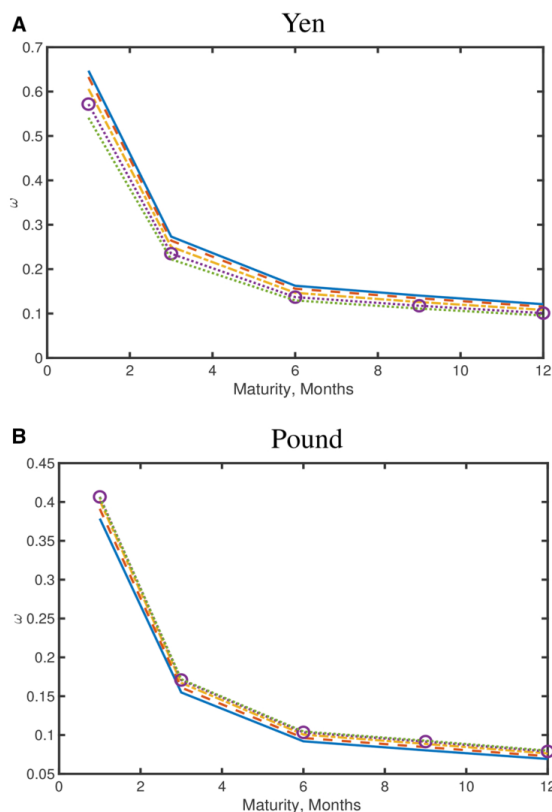
(b) Figure 2: term structure and skew

5.2 Figures 3–4: historical variance estimators

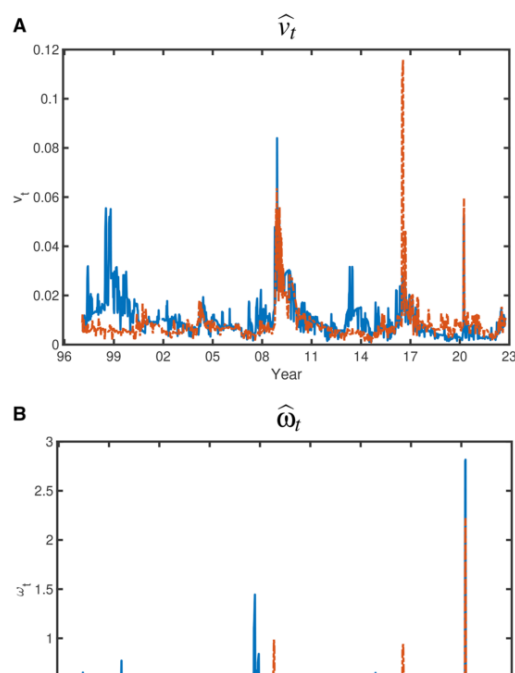
Read:

- Figure 3 shows the average term structure of historical implied-volatility variance estimators.
- The variance estimators are highest at short maturities and decline sharply with maturity.
- Figure 4 shows that return variance and implied-volatility variance estimators spike during crisis periods.
- These historical risk estimates differ across currencies and contracts.

Economic message: The model's risk adjustments are not arbitrary. They reflect actual time-series behavior of the underlying return and each contract's implied volatility.



(a) Figure 3: historical implied-volatility variance



(b) Figure 4: return and implied-volatility variance

5.3 Table 1 and Figure 5: option returns and risk-targeting portfolios

Read:

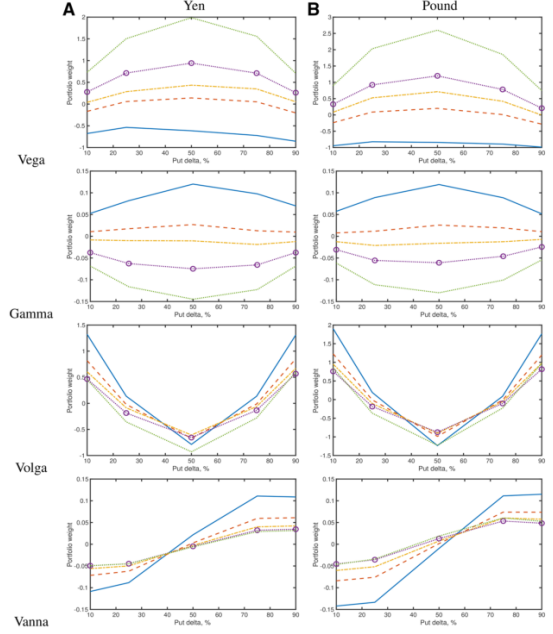
- Table 1 reports delta-hedged excess returns from writing options.
- Mean returns are positive across most moneyness/maturity cells for both currencies.
- Volatility is generally higher for at-the-money options than for deep out-of-the-money options.
- Figure 5 shows that the model's risk-targeting portfolio weights resemble familiar option strategies.
- Vega and gamma portfolios have calendar-spread-like structures. Volga resembles a butterfly, and vanna resembles a risk reversal.

Economic message: The model translates abstract regression coefficients into economically meaningful option portfolios that traders already understand.

Summary behaviors of delta-hedged excess returns for shorting currency options									
Currency	Yen					Pound			
Delta	10	25	50	75	90	10	25	50	75
Maturity	A. Annualized mean excess return, %								
1	0.99	0.86	0.27	0.24	0.34	0.67	0.65	0.25	0.42
3	0.57	0.32	0.10	0.12	0.48	0.56	0.50	0.23	0.31
6	0.50	0.24	0.26	0.18	0.51	0.50	0.45	0.25	0.30
9	0.49	0.28	0.33	0.27	0.45	0.44	0.42	0.31	0.28
12	0.41	0.19	0.38	0.28	0.46	0.40	0.39	0.27	0.34
Maturity	B. Annualized return volatility, %								
1	0.59	1.08	1.41	1.36	1.13	0.59	0.86	1.03	0.77
3	0.58	1.08	1.37	1.26	0.77	0.60	0.88	1.06	0.81
6	0.59	1.08	1.33	1.12	0.72	0.58	0.89	1.08	0.85
9	0.62	1.14	1.39	1.11	0.77	0.63	0.97	1.18	0.95
12	0.63	1.12	1.35	1.10	0.79	0.60	0.95	1.19	0.95

Entries report the summary statistics of the delta-hedged excess returns from writing options across the five delta (in absolute put delta percentages), the five maturities (in months), and the two currencies.

(a) Table 1: delta-hedged excess returns



(b) Figure 5: risk-targeting portfolio weights

5.4 Table 2 and Figure 6: market prices of risk

Read:

- Table 2 summarizes market-price-of-risk estimates for yen and pound options.
- Vega and volga risk prices are positive and statistically strong for both currencies.
- Gamma risk prices are small and weakly negative.
- Vanna risk has different signs across currencies: positive for yen and negative for pound.
- Figure 6 shows substantial time variation in all risk-price estimates.

Economic message: The option surface prices risk dimensions dynamically. The market price of risk is not a constant parameter; it changes over time and differs across risk sources.

Table 2
Summary statistics of the market price of risk estimates

	A. Yen			B. Pound			C. Correlation
	Mean	SD	NW	Mean	SD	NW	%
Vega	2.78	4.12	7.10	3.64	4.48	10.11	22.67
Gamma	-0.04	0.61	-1.07	-0.05	0.57	-1.04	26.67
Volga	1.06	1.23	8.45	1.39	1.28	14.86	14.00
Vanna	0.14	0.40	2.47	-0.18	0.27	-5.01	-20.91

Entries report the summary statistics of the market price estimates on each risk source for yen (panel A) and pound options (panel B). The statistics include the time-series average (Mean), the standard deviation (SD), and the Newey-West t -value on the average (NW). The last column reports the correlation between the market price of risk time series on the two currencies for each risk source.

(a) Table 2: market-price-of-risk summary

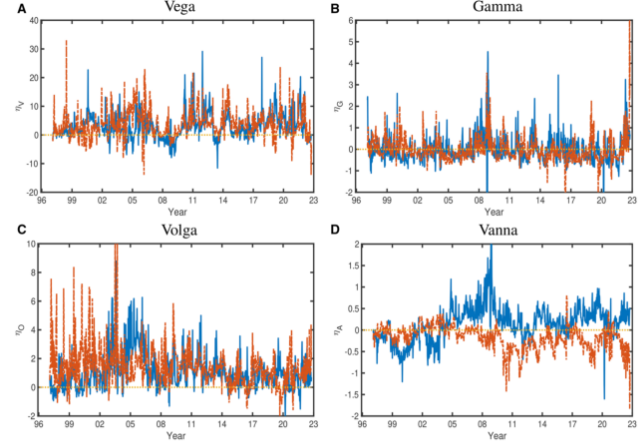


Figure 6
Time series of market price of risk estimates
In each panel, the solid line represents the market price of risk estimates for yen options, the dash-dotted line represents pound options, and the dotted line represents as a zero benchmark.

(b) Figure 6: market-price-of-risk time series

5.5 Table 3 and Figure 7: pricing performance

Read:

- Table 3 reports variance contributions to the cross-sectional pricing regression.
- Gamma explains the largest share of time-decay variation, with 25th–75th percentile contributions around 57–90% depending on currency and percentile.
- Vega is the second-largest contributor, with median contribution around 20%.
- Volga contributes a few percent, and vanna is close to zero in the pricing equation.
- Gross pricing-regression R^2 values are extremely high, with medians around 99.9%.
- Figure 7 shows small implied-volatility pricing errors across the surface, with larger errors at some short-maturity/out-of-the-money corners.

Economic message: The model prices the option surface tightly while still retaining interpretable risk dimensions.

Table 3

Pricing performance and percentage variance contribution from each risk source

VC	A. Yen			B. Pound		
	25	50	75	25	50	75
Vega	3.79	19.78	38.44	6.72	19.60	31.97
Gamma	57.34	74.89	90.48	64.58	76.89	89.31
Volga	3.33	4.28	5.40	2.77	3.52	4.33
Vanna	0.03	0.48	1.14	-0.31	-0.12	-0.00
R^2	99.81	99.89	99.94	99.80	99.90	99.95

Entries report the 25th, 50th, and 75th percentile values of the variance contribution (VC, in percentages) from each risk source to the cross-sectional variation of the option time decay. The last row reports the statistics on the sum of the contributions from all explanatory variables, which represents the gross R^2 of the cross-sectional

(a) Table 3: pricing variance contributions

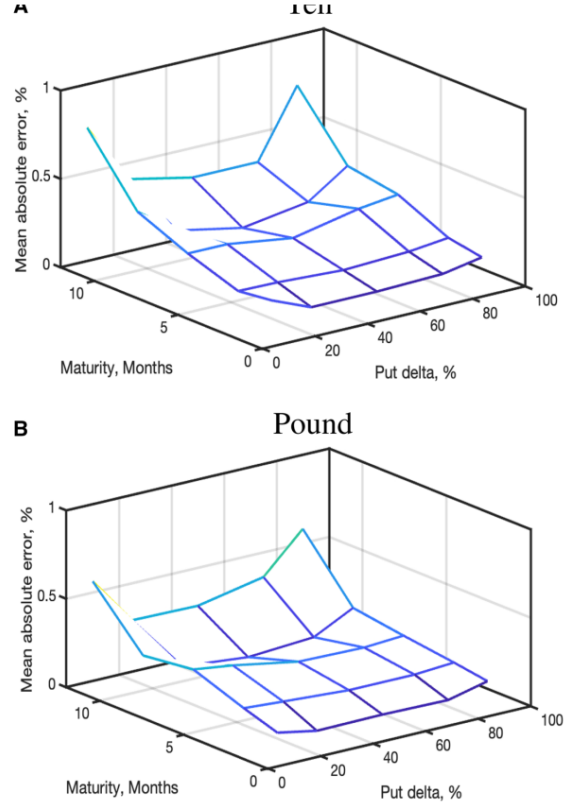


Figure 7

Mean absolute pricing errors on the currency option implied volatility surface

The regression residuals on the options time decay are divided by $B_{SS}^2 S_{t,i}^2$ to convert the errors into t

(b) Figure 7: implied-volatility pricing errors

5.6 Tables 4–6: option return factor model, predictability, and investment performance

Read:

- Table 4 shows that the option return factor model explains cross-sectional option-return variation very well, with average R^2 around 88% for yen and 89% for pound.
- Vega contributes the largest share of realized return variation, around 86–92%.
- The pricing-error factor has positive average excess return and strong Newey–West statistics, consistent with pricing-error reversal.
- Table 5 shows that ex ante market-price-of-risk estimates forecast future risk-targeting portfolio excess returns.
- Forecasting slopes are positive and often close to one; volga slopes are significantly above one.
- Table 6 shows that factor-timing investment improves performance relative to fixed notional short-option portfolios.

Economic message: The model is not only a pricing tool. It also produces risk factors, return forecasts, and implementable option investment strategies.

Table 4
Summary behaviors of the option return factor model

	A. Yen					B. Pound				
	Mean	Volatility	IR	NW	VC	Mean	Volatility	IR	NW	VC
Vega	2.87	6.94	0.41	2.00	86.33	3.77	8.40	0.45	2.17	92.45
Gamma	-0.35	0.71	-0.49	-2.21	-10.37	-0.08	0.57	-0.14	-0.66	-14.64
Volga	2.53	1.96	1.29	6.44	-0.10	3.50	1.87	1.87	10.88	-0.41
Vanna	-0.16	0.42	-0.38	-1.73	9.25	-0.03	0.27	-0.12	-0.62	8.65
Error	1.58	1.03	1.52	7.72	2.92	1.61	1.03	1.56	8.22	3.00
R^2	—	—	—	—	88.03	—	—	—	—	89.05

The table summarizes the estimation results on the option return factor model. For excess return series on each factor portfolio, the statistics include the annualized mean percentage excess return (Mean), the annualized return volatility (Volatility), the annualized information ratio (IR), the Newey-West (NW) t -value on the mean excess return, and the average variance contribution (VC, in percentages) of each factor to the excess return variation.

(a) Table 4: option return factor model

Table 5
Forecasting risk-targeting portfolio excess returns with ex ante market pricing estimates

	A. Yen						B. Pound					
	a_k		b_k		R^2		a_k		b_k		R^2	
Vega	-0.02	(-0.01)	1.04	(3.04)	0.12	0.81	-1.59	(-0.73)	1.47	(3.90)	1.24	1.25
Gamma	-0.31	(-2.04)	0.82	(3.29)	-0.75	1.02	-0.02	(-0.16)	1.31	(6.42)	1.53	3.33
Volga	0.50	(1.00)	1.90	(6.14)	2.92	2.83	0.82	(1.75)	1.93	(7.82)	3.76	3.43
Vanna	-0.29	(-3.00)	0.90	(3.96)	-0.46	1.47	0.11	(1.64)	0.80	(3.91)	-1.01	1.21

Entries report the intercept (a_k), slope (b_k), and the percentage R^2 estimates from the forecasting regression of future annualized excess returns on each risk-targeting portfolio against the ex ante market price of risk estimates on the corresponding risk. In parentheses are the t -statistics, against the null value of 0 for the intercept a_k , and against the null values of 0 and 1, respectively, for the slope coefficient b_k .

(b) Table 5: forecasting risk-targeting returns

Table 6
Investment performance on currency option portfolios

	Yen			Pound		
	Mean	Volatility	IR	Mean	Volatility	IR
<i>A. Fixed notional short investment</i>						
Vega	0.23	0.50	0.46	0.16	0.48	0.33
Gamma	-0.15	0.39	-0.38	-0.08	0.33	-0.24
Volga	0.21	0.20	1.07	0.21	0.13	1.62
Vanna	-0.07	0.32	-0.22	-0.05	0.21	-0.26
Market portfolio	0.35	0.90	0.39	0.38	0.72	0.54
<i>B. Factor timing investment</i>						
Vega	0.61	0.67	0.91	0.46	0.63	0.72
Gamma	0.13	0.62	0.21	0.22	0.52	0.42
Volga	0.11	0.11	1.06	0.13	0.10	1.34
Vanna	0.39	0.48	0.83	0.14	0.19	0.72
Risk premium harvesting	0.31	0.26	1.19	0.24	0.25	0.95

Entries report the summary statistics of the investment returns on different currency option portfolios, including the annualized mean return, the annualized return volatility, and the annualized information ratio. The two panels are for two investment strategies. Panel A invests a fixed \$100 notional short in each risk-targeting portfolio at all days, with the last row summarizing the return behavior for shorting an equally weighted market portfolio of all option contracts. Panel B performs market timing investment in each risk-targeting portfolio at each date in proportion to the corresponding market price of risk estimate on that date. The last row summarizes the equal-weighted average return behavior from the factor-timing investments in all four risk-targeting portfolios as a way of harvesting the risk premium from the four risk dimensions. The market timing investments are normalized

Figure 7: Table 6: investment performance on currency option portfolios

6 Short summary

- Wu and Zhang propose a top-down linear option pricing model.
- The model starts from observed option contracts and BMS sensitivities, not from a full underlying risk-neutral dynamics.
- Each option's P&L is decomposed into theta, delta, vega, gamma, volga, and vanna terms.
- After delta hedging, the key risk sources are vega, gamma, volga, and vanna.
- Historical moment estimators measure the risk magnitude for each contract.
- The central assumption is common market pricing per unit of each estimated risk source across option contracts.
- A cross-sectional regression of time decay on risk-adjusted exposures identifies the market price of risk at each date.
- The residuals are pricing errors relative to common risk pricing and can generate statistical-arbitrage interpretations.
- The empirical application uses OTC options on yen and pound from 1996 to 2022.
- The model prices the currency option surfaces with extremely high cross-sectional R^2 and low implied-volatility errors.
- The risk-price estimates are time varying; vega and volga risk prices are strongly positive on average.
- The model-implied risk-targeting portfolios resemble calendar spreads, butterflies, and risk reversals.
- The same risk-adjusted exposures explain realized delta-hedged option returns.
- Market-price-of-risk estimates forecast future returns of matching risk-targeting portfolios.
- Factor-timing strategies based on these estimates improve risk-adjusted performance.

7 Conclusion

7.1 Core conclusion

The paper shows that option contracts can be priced coherently without specifying a full bottom-up dynamics. The key is to decompose each contract into decentralized risk exposures and require common pricing per unit of estimated risk across contracts.

7.2 Economic intuition

Each option has its own risk behavior. A short-dated out-of-the-money option and a long-dated at-the-money option should not be forced to share the same implied-volatility dynamics. But if both carry vega risk, gamma risk, volga risk, or vanna risk, investors should price each unit of that risk consistently. The model uses this consistency condition to identify market prices of risk through a simple cross-sectional regression.

7.3 Takeaway

The paper provides a bridge between option valuation and option risk management. BMS sensitivities become not just hedging Greeks but the basis for a linear asset-pricing model of option surfaces and option returns. The main message is that common pricing of decentralized risk can be more useful than common dynamics when option risks are heterogeneous across contracts.